

RESEARCH ARTICLE

Leveraging state-of-the-art AI models to forecast wind power generation using deep learning

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Abstract

In this paper, we present a novel approach for forecasting weather variables that are not currently available in many state-of-the-art AI models. A variable not found in most models is the 100-m wind speed, which is commonly used in the energy sector to predict the power generated by wind turbines. We trained a convolutional neural network model on 12 years of ERA5 data to instantaneously predict the 100-m wind speed based on a subset of variables found in the ECMWF-AIFS forecast. We evaluated our model with 2020 ERA5 data and achieved an average 100-m wind speed RMSE of 0.18 m/s, outperforming the wind profile power law method with an RMSE of 0.63 m/s. Using the AIFS output as input to our trained model, we generated 10-day 100-m wind speed forecasts without requiring autoregressive steps, significantly reducing computational costs. We compared our predictions with the ECMWF-IFS forecast using the ECMWF analysis as ‘ground truth’ and showed greater accuracy at longer lead times. Additionally, we produced power generation forecasts for onshore and offshore wind farms across the United Kingdom, with improvements over the IFS after a lead time of 3 days. We also showed that our model exhibits spatial and temporal coherence between local predictions and discussed the common limitation of over-smoothing in the predictions of AI models.

KEYWORDS

100-m wind speed, artificial intelligence, weather forecasting, wind power generation

1 | INTRODUCTION

The use of AI to produce weather forecasts that rival the accuracy of current NWP models has been a popular research domain in recent years. Notable models include AIFS (Lang et al., 2024), GraphCast (Lam et al., 2023), Pangu-Weather (Bi et al., 2023) and FourCastNet (Kurth et al., 2023), all producing reliable medium-range forecasts

at a fraction of the computational cost required to run NWP models. Despite the success of AI models for weather prediction, the chosen output variables are limited to produce a tractable model, given the current standards in computational power, resulting in a lack of AI-based forecasts for certain applications.

Variables that are not currently forecast in most AI models are the u and v components of the 100-m wind

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speed, which are commonly used in the energy sector to predict the power generated by wind turbines. With limited AI-based models, energy companies still rely on NWP methods to obtain 100-m wind speed predictions, which is a time-consuming and costly process that does not take advantage of historical data to improve forecasts. Previously, machine learning models have been created to produce 100-m wind speed forecasts using historical data; however, these tend to perform poorly at longer lead times and are often limited to lower-resolution data due to computational constraints (Xie et al., 2023). Only with recent developments in deep learning methods, such as transformers (Vaswani et al., 2017) and GCNs (Kipf & Welling, 2017), are AI models able to accurately capture these long-term dependencies at higher resolutions; however, these require significantly more computational resources to train.

In this paper, we present a novel method to forecast additional weather variables, such as the 100-m wind speed components, by leveraging the output of a pre-existing AI model. We take advantage of the performance and efficiency of convolutional neural networks (CNNs) to produce a relatively simple model that accurately forecasts 100-m wind speeds over Europe using the predictions from a state-of-the-art AI model at each time step as our model's inputs. With our method, we are able to predict additional variables not considered in pre-existing AI models, without the computational burden of training our own autoregressive model to forecast the 100-m winds from an initial state. Training autoregressive models involves initialising from the previous prediction for each step in the forecast, which results in multiple runs of the model per iteration. Whilst creating an AI model in this fashion may produce superior 100-m wind speed forecasts, training them typically requires many GPUs over long time periods, incurring a large computational cost. Instead, we chose to leverage the existing AIFS forecast for our model input due to the commercial licensing terms and the availability of the data from the ECMWF Data Store. Following this, our 100-m wind speed predictions are used to forecast the power generated by onshore and offshore wind farms across the United Kingdom and are compared to the current ECMWF NWP model (IFS) and ECMWF analysis data to evaluate the performance of our model.

2 | METHODOLOGY

2.1 | Overview

In this section, we present our model to predict the 100-m wind speeds over a European latitude-longitude grid and the methods used to develop it. We use AI-based

approaches to create an algorithm that can accurately predict instantaneous 100-m wind speeds by training on 12 years of ERA5 data at hourly temporal resolution and $0.25^\circ \times 0.25^\circ$ grid spacing. Our proposed model is a CNN based on the UNet (Ronneberger et al., 2015) architecture created for medical image segmentation, which has been previously shown to be successful for weather forecasting on a global grid (Weyn et al., 2020). By using the operational forecast from the state-of-the-art AIFS model as inputs to our ERA5-trained model, we produce a 10-day forecast of 100-m winds without training an autoregressive model, which surpasses the IFS in RMSE at longer lead times. Additionally, we explore the application of our model to predict the power generation of onshore and offshore wind farms across the United Kingdom.

2.2 | Data preparation

We used 14 years of the complete ERA5 global atmospheric reanalysis (Hersbach et al., 2020) data were downloaded using the Copernicus Climate Data Store API for model training and preliminary evaluation. For training, we used the years 2010 through 2019, 2022 and 2023, and for testing and validation, we used the years 2020 and 2021, respectively. We used a recent range of years for training to model a more current state of the earth's climate and to reduce computational cost. We note that this choice of data split resulted in leakage during the preliminary evaluation of 2020 ERA5 data; however, excluding 2021–2023 would be an unnecessary reduction in skill as our model forecasts using AIFS data, which was only made available in 2024. Additionally, testing on 2020 data is the current reference standard in the AI community and is also used in benchmark datasets such as WeatherBench 2 (Rasp et al., 2023). ECMWF analysis data for 2020 was also used for model testing. The data contains surface and pressure variables at an hourly resolution on a 0.25° grid over Europe, with a latitude range of $[30, 75]$ and a longitude range of $[-30, 45]$.

Our model uses a subset of variables predicted by the AIFS model to produce an instantaneous prediction of the 100-m wind speed components at a given time t for all latitude-longitude points. To reduce computational cost, we use pressure variables at 500, 600, 700, 850, 925 and 1000 hPa, as they are more physically relevant in predicting the 100-m winds than the higher levels. Additionally, we include six static variables in the model inputs to give context about the terrain over which the wind is predicted. The concatenation of these variables results in an input of size $50 \times 181 \times 301$, corresponding to variable $\times$ latitude $\times$ longitude. Table 1 summarises the input and output variables used by our model.

TABLE 1 Variables used for the model's inputs and outputs.

Variable	Short name	Type	Role
Mean sea level pressure	msl	Surface	Input
Skin temperature	skt	Surface	Input
Surface precipitation	sp	Surface	Input
Total precipitation	tp	Surface	Input
10 m <i>u</i> wind component	u10	Surface	Input
10 m <i>v</i> wind component	v10	Surface	Input
2 m temperature	t2m	Surface	Input
2 m dewpoint temperature	d2m	Surface	Input
Geopotential	z	Pressure	Input
<i>u</i> wind component	u	Pressure	Input
<i>v</i> wind component	v	Pressure	Input
Specific humidity	q	Pressure	Input
Vertical velocity	w	Pressure	Input
Temperature	t	Pressure	Input
Geopotential at surface	z	Static	Input
Land-sea mask	lsm	Static	Input
Slope of sub-gridscale orography	slor	Static	Input
Angle of sub-gridscale orography	anor	Static	Input
Anisotropy of sub-gridscale orography	isor	Static	Input
Standard deviation of sub-gridscale orography	sdor	Static	Input
100-m <i>u</i> wind component	u100	Surface	Output
100-m <i>v</i> wind component	v100	Surface	Output

Note: The 'Variable' and 'Short Name' columns correspond to the long and short names as given in the ERA5 documentation. The 'Type' column indicates whether the variable is invariant in time (static), is time-dependent at a single level (surface), or time-dependent at the six included pressure levels (pressure). The 'Role' column indicates whether the variable is an input or an output of the model.

As a preprocessing step, we standardise the data to zero mean and unit variance using the per-variable mean and standard deviation calculated across the training set. The model predicts standardised components of the 100-m wind speed, which are scaled back to their original distribution during inference using the training set mean and standard deviation. This was done to enhance numerical stability during training and prevent issues caused by the varied magnitudes of the input variables.

In addition to ERA5, we also downloaded AIFS and IFS forecasts and ECMWF analysis data from March 2024 to November 2024 to evaluate the forecast capability of our model. The IFS forecasts and ECMWF analyses were regridded from their native 0.1° grid to 0.25° to match the AIFS and ERA5 datasets. We use this range of data as the AIFS model has only been operational since late February 2024, and there is no back-dated data available. This means that the initial results below are not representative of the entire year, an issue when testing any new model that does not have associated reforecast data. By using the 10-day forecast produced by the AIFS model as inputs for

our model, we produce a 100-m wind speed forecast without using autoregressive steps. Using the ECMWF analysis as 'ground truth', we compared the 10-day forecasts produced by our model and the IFS for wind speed and power generation at onshore and offshore wind farms across Britain. Despite our model being trained on ERA5 data, we chose to evaluate it against the ECMWF analysis as the AIFS and IFS models use the ECMWF initialisation to generate their forecasts. Since our model will be using the AIFS predictions as inputs to produce operational forecasts, we will verify its performance to an operational 'ground truth' as opposed to the ERA5 reanalysis data. Figure 1 shows a flowchart that outlines the stages of the analysis in which each of the 'ground truth' data sets are used and their corresponding purpose.

2.3 | Model architecture and training

Our model is a fully convolutional encoder-decoder network based on the UNet architecture designed for

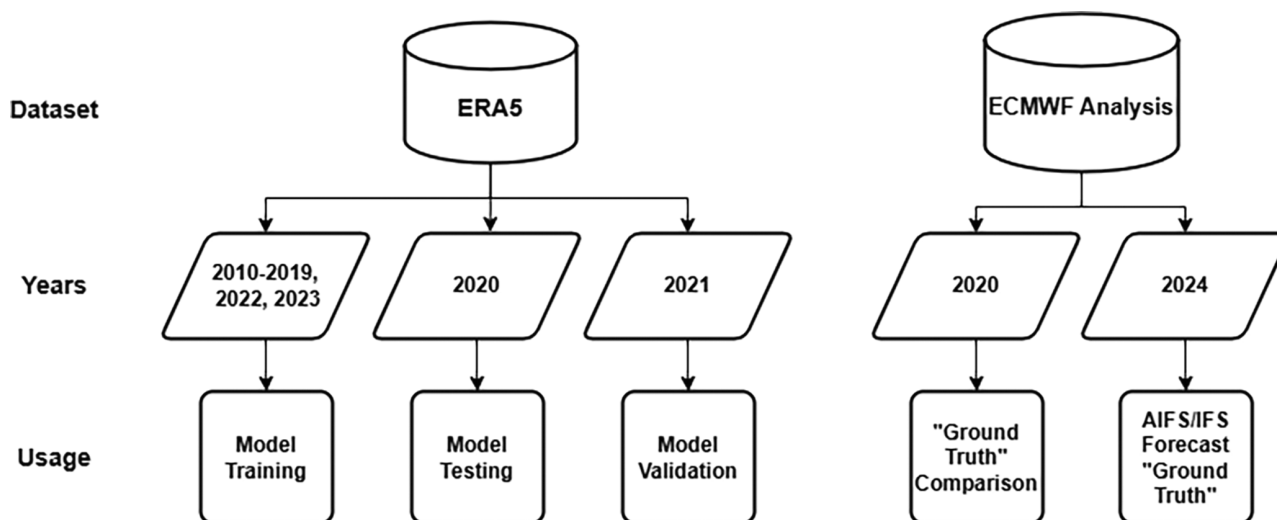


FIGURE 1 Flowchart illustrating how each of the ‘ground truth’ data sets are used throughout the analysis. The top row of the flowchart shows the two data sets, the middle row indicates what years are used and the bottom row describes the purpose of the data.

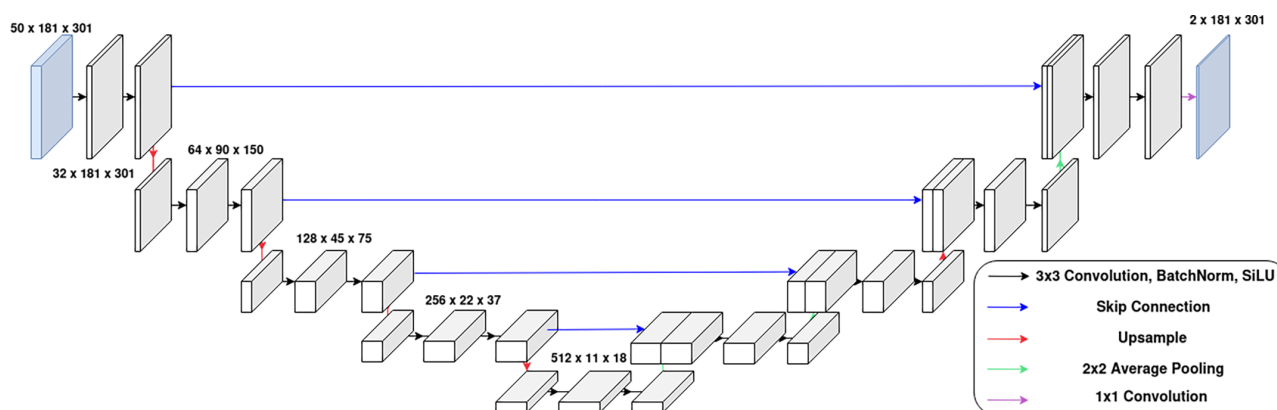


FIGURE 2 Schematic diagram of the model used to predict the 100-m wind speed u and v components. The model is based on the UNet architecture, a fully convolutional encoder-decoder model used for image segmentation. The blue tiles represent the input and output grids, and the grey tiles show the latent representations of the data as it is processed by the model. The numbers next to the tiles describe the shape of the data at that step, corresponding to channels $\times$ latitude $\times$ longitude. The black arrows represent a 3×3 convolution followed by a SiLU activation function and batch normalisation, the red arrows illustrate a 2×2 pooling operation and the green arrows represent a 2×2 upsampling step. The blue arrows show skip connections, and the purple arrow is a 1×1 convolution.

biomedical image segmentation. We predict the u and v components of the 100-m wind speed by grid point using an input state of 50 variables at time t . The convolutional layers in the network extract features using a sliding ‘filter’ over the input space to process information from neighbouring grid points and map local relationships between variables. Additionally, the weight-sharing mechanism of CNNs allows the model to learn long-range dependencies across the grid space, making it an attractive option for geographical data. The model predictions are compared to a ‘ground truth’ 100-m wind speed grid at the input time t , where the error gradients are backpropagated to learn optimal weights for the convolutional filters.

Figure 2 shows a schematic diagram of our proposed model. The encoder (left side) consists of a sequence of 3×3 convolutional layers with increasing filters to extract hierarchical representations of the input data. Each convolution is followed by a batch normalisation (Ioffe & Szegedy, 2015) layer to prevent internal covariate shift and reduce overfitting, and a sigmoid linear unit (SiLU) activation function (Elfwing et al., 2017) to introduce non-linearity. Every second convolution in the encoder is followed by a 2×2 average pooling operation to reduce the dimensionality of the current state, whilst retaining the essential information. The decoder (right side) essentially mirrors the encoder but instead replaces the pooling operations with bilinear upsampling to

incrementally return to the input spatial dimension of 181×301 . Skip connections are also added after each upsampling step by concatenating the output at the respective encoder step to ensure maximum information flow throughout the network. The final layer is a 1×1 convolution to map the feature channels to the number of output variables, in our case 2. This architecture results in a model with 5,512,130 trainable parameters, a relatively small number by modern standards.

Our model is trained to minimise a latitude-weighted mean-squared-error loss function between the target y and output $\hat{y}$ calculated per mini-batch of samples, given by

$$\mathcal{L}(y, \hat{y}) = \frac{1}{2} \sum_{k \in \{u, v\}} \frac{1}{N_b} \sum_b \frac{1}{N_i} \sum_i \frac{1}{N_j} \sum_j w_i \left(y_{k,i,j}^b - \hat{y}_{k,i,j}^b \right)^2, \quad (1)$$

where

- k represents the wind speed component,
- b denotes the sample index in a mini-batch of size N_b ,
- i is the latitude index in a grid with N_i points,
- j is the longitude index in a grid with N_j points,
- w_i is the per-latitude weighting given by the cosine of the latitude.

The weighting accounts for the decreasing grid sizes as the latitude increases. We trained our model using the AdamW (Loshchilov & Hutter, 2019) variant of the stochastic gradient descent algorithm with an initial learning rate of 5×10^{-5} and a weight decay of 10^{-5} . A scheduler was implemented to reduce the learning rate by a factor of 10 if the validation loss did not improve by 10^{-4} for 5 epochs. The model was trained for 100 epochs using a batch size of 32 on a single Nvidia 3060 12 GB GPU and took 30 h to complete.

2.4 | Wind power generation

Following the analysis performed by Bloomfield et al. (2022), the predicted 100-m wind speeds from our model were used to produce 10-day power generation forecasts for 1120 onshore and 55 offshore wind farms across the United Kingdom. To accomplish this, we acquired data from The Wind Power (The Wind Power, 2024) containing the location, hub heights and total power output of wind farms in the United Kingdom. We selected a subset of farms within Britain and performed a bilinear interpolation from the 0.25° grid to determine an approximate 100-m wind speed at each farm's exact location. The wind

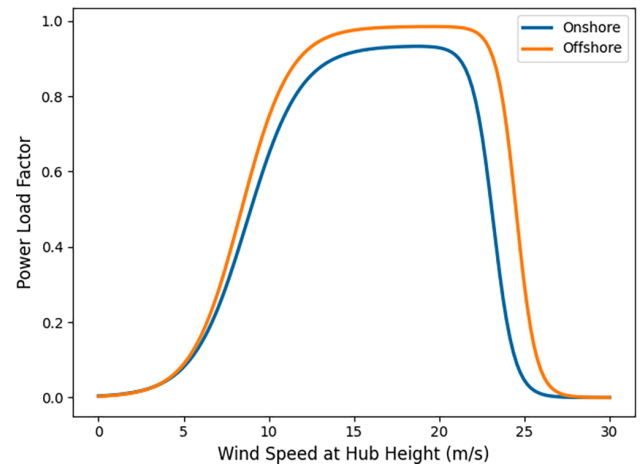


FIGURE 3 UK onshore and offshore power curves taken reproduced from Bloomfield et al., (2022) (CC 4.0), who cites a National Grid report (National Grid, 2019). The curves convert wind speed into a power output at various wind farms across the United Kingdom.

speeds were then scaled to the turbine hub height using the wind profile power law:

$$U_1 = U_2 \left(\frac{Z_1}{Z_2} \right)^\alpha, \quad (2)$$

where Z_1 is hub height, Z_2 is the wind speed height (100 m), U_1 and U_2 are the wind speeds at the respective heights, and α is an empirically derived stability coefficient which we set to $\frac{1}{7}$ for neutral conditions. The wind speeds were then passed through a generic onshore or offshore power curve taken from a National Grid report (National Grid, 2019) to produce a load factor at input time t per farm. Figure 3 illustrates the onshore and offshore power curves. This process was performed using our AIFS 100-m wind speeds and the IFS 100-m wind speeds, both using the ECMWF analysis as 'ground truth' for evaluation.

3 | RESULTS AND DISCUSSION

3.1 | 2020 analysis

The 2020 ERA5 testing dataset was used to evaluate the instantaneous predictive performance of the proposed model at hourly resolution. For each time point in the dataset, the variables listed in Table 1 were used as inputs to the model to produce the 100-m wind components across the European grid, which were then used to calculate the magnitude of the 100-m wind speed. The year-averaged latitude-weighted RMSE between the ERA5

‘ground truth’ and the model predictions was determined using

$$RMSE(y, \hat{y}) = \sqrt{\frac{1}{N_t} \sum_t \frac{1}{N_i} \sum_i \frac{1}{N_j} \sum_j w_i (y_{ij}^t - \hat{y}_{ij}^t)^2}, \quad (3)$$

with t indicating the time index out of N_t time steps, for the u and v components, and the magnitude of the 100-m wind. The RMSEs were 0.16, 0.16 and 0.18 m/s, respectively, suggesting our model can predict the 100-m winds with high accuracy.

In order to check that our CNN model outperforms a commonly used baseline (Bloomfield et al., 2022; Ramon et al., 2019), we also modelled the 100-m wind speeds using the wind profile power law in Equation (2) with the 10 m wind speeds as U_1 and Z_1 equal to 10 m. We fit Equation (2) to the ERA5 training data to determine a value for α at each grid point, which acts as a proxy for the surface roughness, the zero-plane displacement and the stability of the atmosphere. The wind power profile model was then evaluated on the 2020 hold-out set, yielding an average RMSE of 0.63 m/s and demonstrating that our CNN model performs significantly better in predicting the 100-m wind speeds. There are more sophisticated methods of calculating 100-m wind speeds, but that is not the focus of this paper.

To understand where the errors arise in our model and in the wind profile power law, heat maps displaying the time-averaged RMSE at each grid point were generated, as shown in Figure 4. The largest errors in our CNN model predictions (left) generally correspond to high-altitude areas

such as mountain ranges and coastlines. Additionally, the offshore RMSE is significantly lower than on land, likely caused by the minimal variations in topology and surface roughness compared to onshore, similarly for the wind profile power law (right). The predictions from the wind profile power law are also better near coastlines and across mountain ranges, which is thought to be due to the sea and mountain breezes that can dominate these regions. Large amounts of smoothing were observed in the wind profile power law predictions, highlighting a limitation of using simple approximations to model the wind speeds.

We compared and visualised the grid-averaged wind speed distributions across 2020 for the ERA5 and ECMWF analysis datasets in Figure 5 to illustrate the differences between the two ‘ground truth’ datasets. The data are compared using a violin plot where each side is one of the compared distributions that are smoothed using a kernel density estimation. The predicted distribution for each violin plot is the same, generated using ERA5 data as inputs to the model. As expected, the predicted distribution matches almost exactly with the ERA5 100-m wind speed, shown in the left plot, as this was the dataset used to train the model. The ECMWF analysis distribution shown in the right plot also matches in shape with the model outputs; however, there is a bias in the wind speed predictions, caused by the slight differences between the two datasets. Since the AIFS is also trained on ERA5 data and the ECMWF initialisation is used to produce operational forecasts, this bias will need to be corrected for the 100-m wind speed predictions. The difference calculated between the predictions and the EC analysis 2020 data is used to correct the operational AIFS 100-m wind speed forecasts generated by our model.

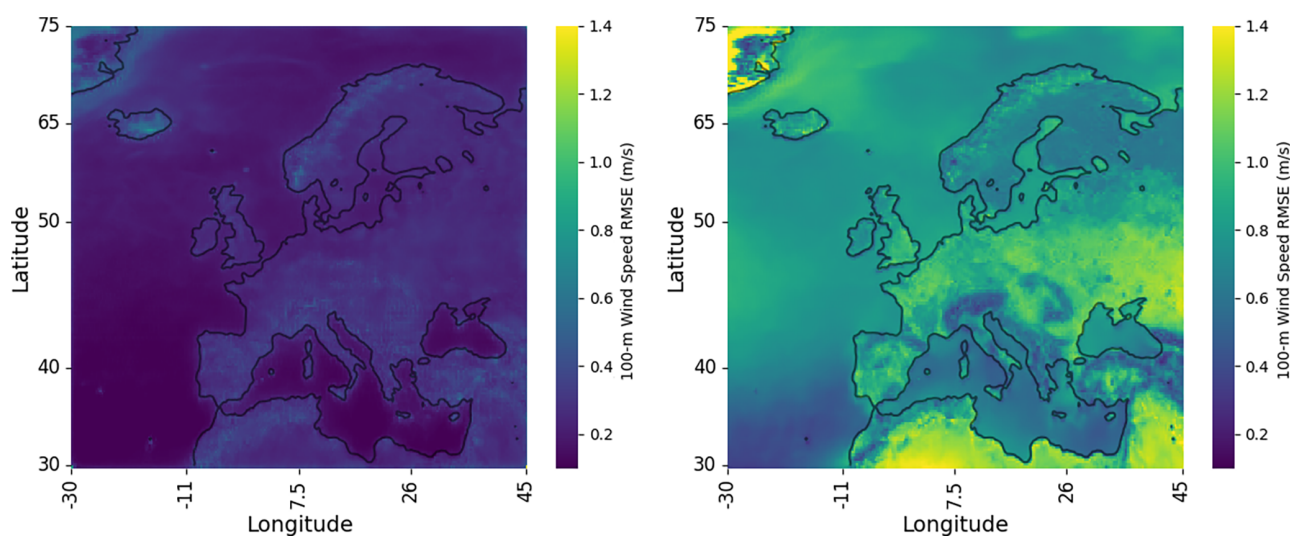


FIGURE 4 Heat map of latitude-longitude grid points where the hue represents the time-averaged wind speed RMSE using the ERA5 2020 dataset as ‘ground truth’. The left plot shows the 100-m wind speed RMSE predicted by our CNN model, and the right plot shows the results from the wind profile power law. The static ERA5 land-sea mask is overlaid on the heat map, as shown by the black lines.

3.2 | 2024 100-m wind speed 10-day forecast

Our model was fed the AIFS forecasts initialised at 00 UTC (00Z) each day to produce a 10-day forecast for the u and v components of the 100-m wind speed at 6-h intervals. Following this, the magnitude of the wind

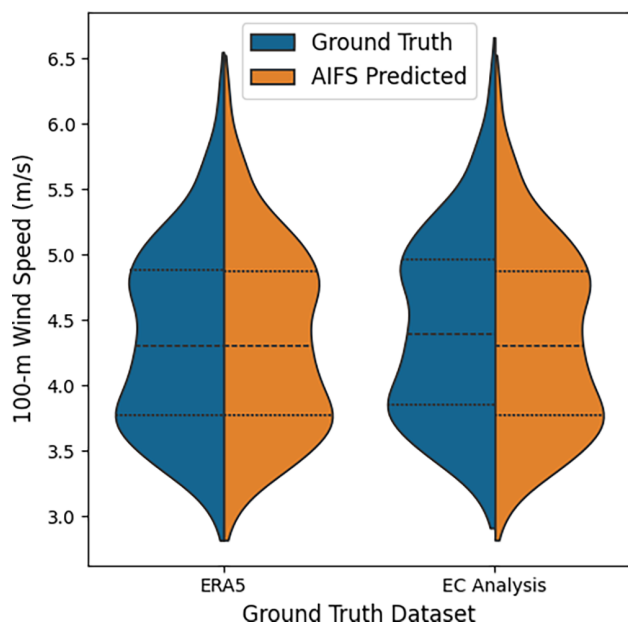
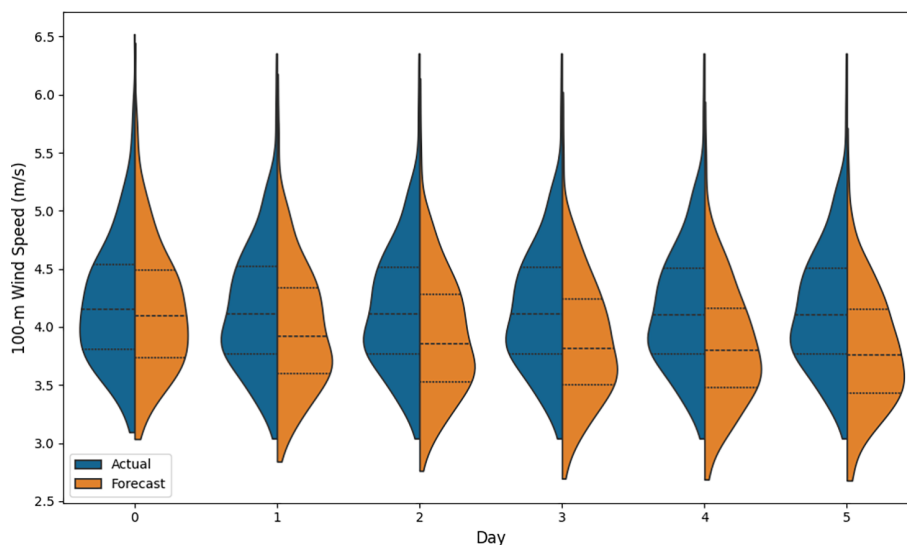


FIGURE 5 Violin plots comparing the predicted (orange) 100-m wind speed distribution with the 2020 ERA5 and ECMWF analysis 'ground truth' datasets (blue). The predictions are generated using ERA5 data as model inputs and are the same for both plots. The left plot compares the predictions to the actual ERA5 distribution, and the right plot compares to the ECMWF analysis. The dashed line represents the mean and the dotted lines describe the quartiles.

speed was calculated from the components for each data point. To compare the accuracy of our forecast, the 100-m wind speeds from the ECMWF analysis at 00Z, 06Z, 12Z and 18Z from March 2024 to November 2024 were used as the 'ground truth'. Figure 6 shows violin plots comparing the distribution of the actual grid-averaged wind speeds across 2024 to the first 5 days of the 100-m wind forecast. The wind speeds in the figure are adjusted using the bias calculated between the 2020 ERA5 predictions and the ECMWF analysis data, and averaged on a daily basis, that is, day 1 is the average of the 6, 12, 18 and 24-h steps, except for the 0-h step. At step 0 the AIFS outputs are essentially the 00Z ECMWF analysis values and so we observed similar actual and forecast distributions, reinforcing the predictive capabilities of our model. At the subsequent time steps, the shape of the forecast distribution closely matches the actual data; however, a 'drift' was observed, resulting in increasingly underforecast wind speeds over time. We suspect this error arises due to the difference between the ERA5 and ECMWF analysis datasets, as noted in the previous section. As the AIFS model produces forecasts using autoregressive steps, these small errors accumulate over time, resulting in the observed shift, highlighting a limitation when relying on the accuracy of another model for our predictions. Due to these increasing errors, the wind speed adjustment only accurately corrects the bias in the 0-h step as these values are not produced via autoregression.

As done on the 2020 analysis, we visualised the RMSE by grid point for the first 5 days in the forecast to identify how the errors evolve, as shown in Figure 7. At the earlier lead times, the errors were larger in areas of higher altitude and across the southern European Mediterranean and Greenland coasts, similar to the 2020 results. Over time, the errors in the prediction become dominated

FIGURE 6 Violin plots comparing our AIFS-based predictions (orange) with the ECMWF analysis 'ground truth' (blue) for the 100-m wind speed from March 2024 to November 2024. Each violin plot represents the wind speed distribution for a day in each forecast, that is, day 1 is the distribution of the 6, 12, 18 and 24-h steps. The dashed line represents the mean and the dotted lines describe the quartiles.



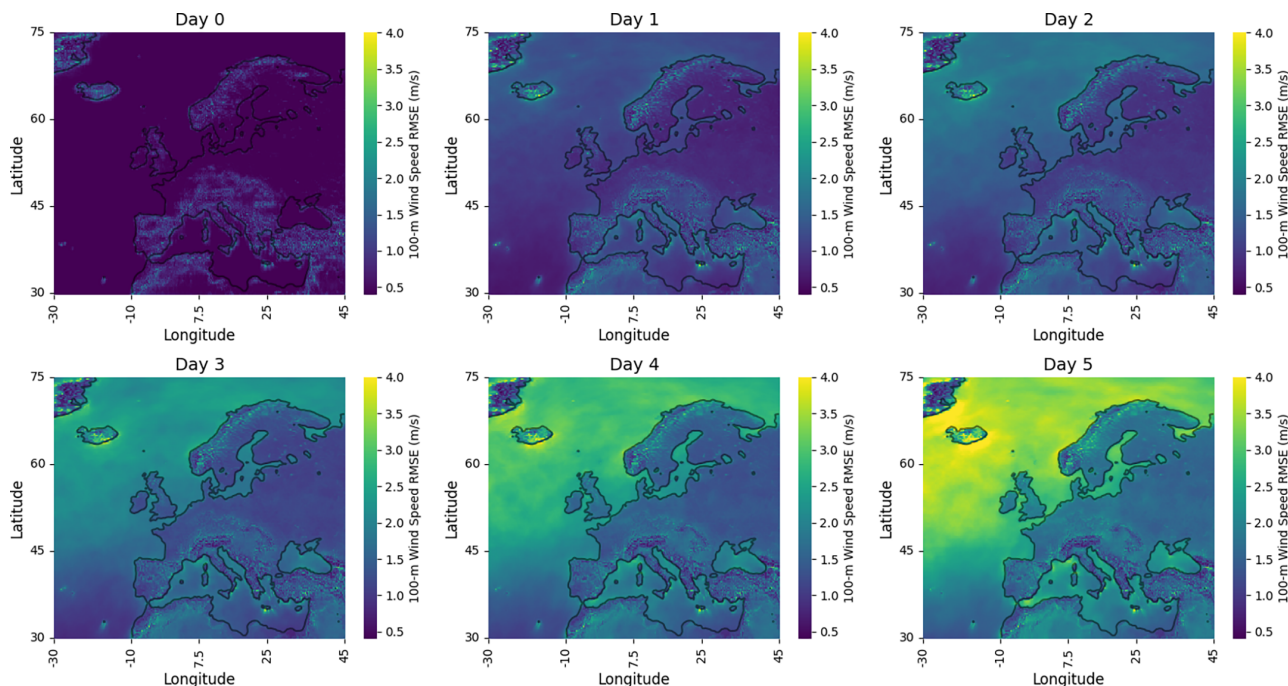


FIGURE 7 Heat map of latitude-longitude grid points where the hue represents the time-averaged 100-m wind speed RMSE from March 2024 to November 2024. Predictions were generated with our model using the AIFS forecast as inputs and using the ECMWF analysis as ‘ground truth’. Each map represents the average wind speed RMSE for a day in each forecast, that is, day 1 is the average of all the 6, 12, 18 and 24-h steps. The static ERA5 land-sea mask is overlaid on each heat map, as shown by the black lines.

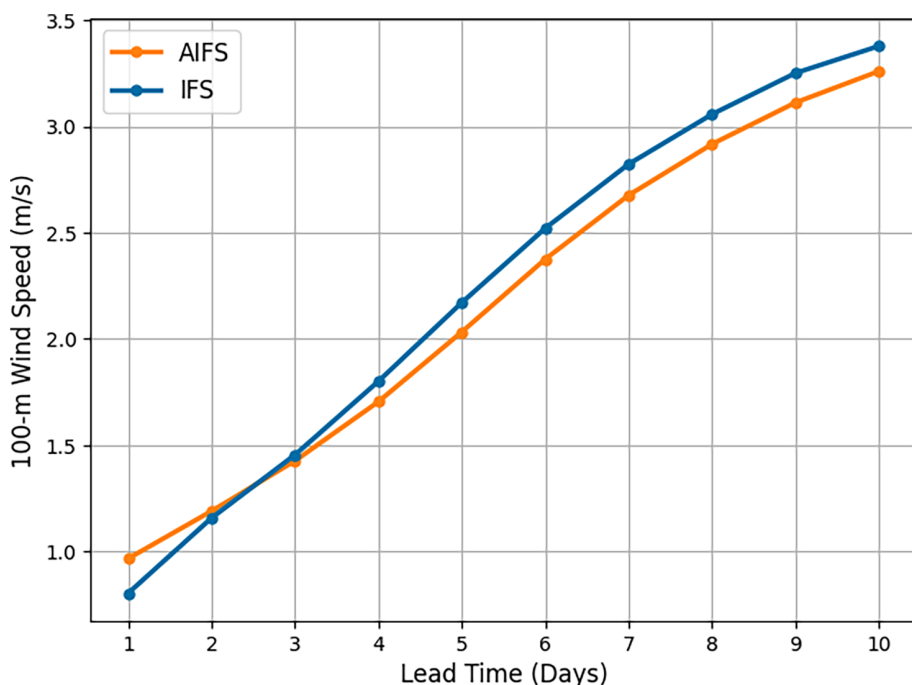


FIGURE 8 100-m wind speed RMSE by forecast lead time in days for our AIFS-based predictions (orange) and IFS (blue) model from March 2024 to November 2024. The ECMWF analysis was used as the ‘ground truth’. The RMSE is averaged over the grid with a latitude weighting and then averaged by day in the forecast, that is, day 1 is the average of all the 6, 12, 18 and 24-h steps.

by the North Atlantic region, particularly off the eastern coast of Greenland, as expected due to the noted bias. In contrast to what was observed at the earlier lead times and in the 2020 analysis, the wind speed error at day 5 and above is lowest across the higher altitude regions.

In analysing the 10 m wind speed forecasts generated by the AIFS, we notice a similar error; these errors arise from the AIFS as opposed to our proposed model.

We also compared the forecast capabilities of our model to the IFS (without any bias correction). Figure 8

illustrates the RMSE of the AIFS and IFS predictions when compared to the ECMWF analysis, which was calculated using Equation (3) across 2024 and averaged over each day. We found that for the first few days, our model did not surpass the accuracy of the IFS, likely due to the bias in the predictions; however, at day 3 and beyond, we achieved superior predictive capabilities, shown by a lower average RMSE. We additionally visualised the RMSE per 6-h time step to identify any diurnal cycles in our predictions; however, nothing of significance was found, so it was omitted from this report. For the longer forecast times, one would expect an ensemble mean to outperform a deterministic run: we await the availability of AI ensemble data to evaluate the improvement ensembles can provide here and expect the deterministic AIFS skill to represent a lower limit.

3.3 | 2024 UK power generation 10-day forecast

With the predicted 100-m wind speeds from 2024, we created 10-day forecasts for the power generated by onshore and offshore farms across the United Kingdom. The steps outlined in Section 2.4 were followed for the AIFS and IFS predictions and the ECMWF analysis data to produce load factors for each farm at each time step. For both onshore and offshore wind farms, we observed closely matching distributions at the earlier lead times with a slight drift as the forecast rolled out, as expected due to the bias in the predictions.

As done previously, we compared our AIFS wind power forecast to the IFS predictions. In Figure 9, the load factor RMSE for the AIFS and IFS models at the onshore and offshore wind farms is shown. At the earlier forecast days (<3 days) the IFS beats our model in predicting the power output for both the onshore and offshore farms. Similar to the behaviour observed in the wind predictions, our model shows improved capabilities than the IFS forecast on day 3 and later, with a lower RMSE at all steps. Notably, the offshore predictions contribute more to the overall error, particularly at the later lead times, as observed in our wind speed forecast in Figure 7. Further research could explore using separate models trained for onshore and offshore predictions to reduce this error.

In addition to the ECMWF analysis, we also compared our forecasts to actual UK power generation data downloaded from the BMRS ELEXON portal (ELEXON, 2024). Total UK wind generation is currently a combination of intermittent generation from operational wind farms and sporadic generation from recently constructed wind farms being tested. We therefore chose a single wind farm, Kilgallioch, in western Scotland, with known installed capacity, to give an accurate proof of concept for power forecasting with our model. Figure 10 illustrates the daily power generation RMSE of our model and the IFS when compared to the actual power generation at Kilgallioch in 2024. In line with our previous analyses, we find that our model performs worse at earlier lead times due to the forecast error arising from the different 'ground truth' data and outperforms the IFS at days 4 and above. Our AI-

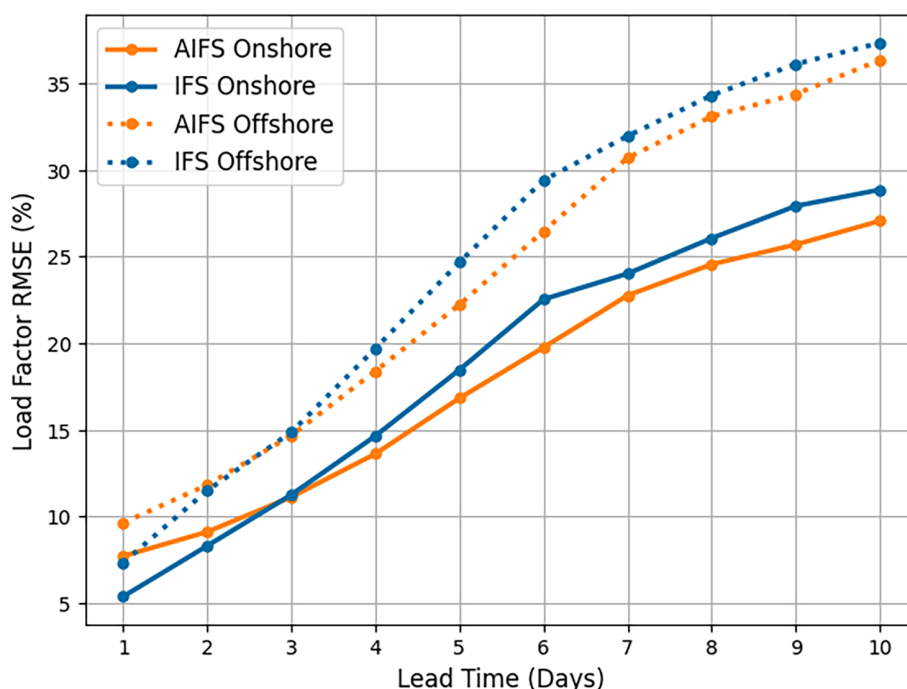


FIGURE 9 RMSE for the 2024 power generation forecasts generated from our model (orange) and the IFS (blue) from March 2024 to November 2024. The ECMWF analysis was used as the 'ground truth'. The RMSE is averaged over each onshore (solid) and offshore (dashed) UK wind farm and then averaged by day in the forecast, that is, day 1 is the average of all the 6, 12, 18 and 24-h steps.

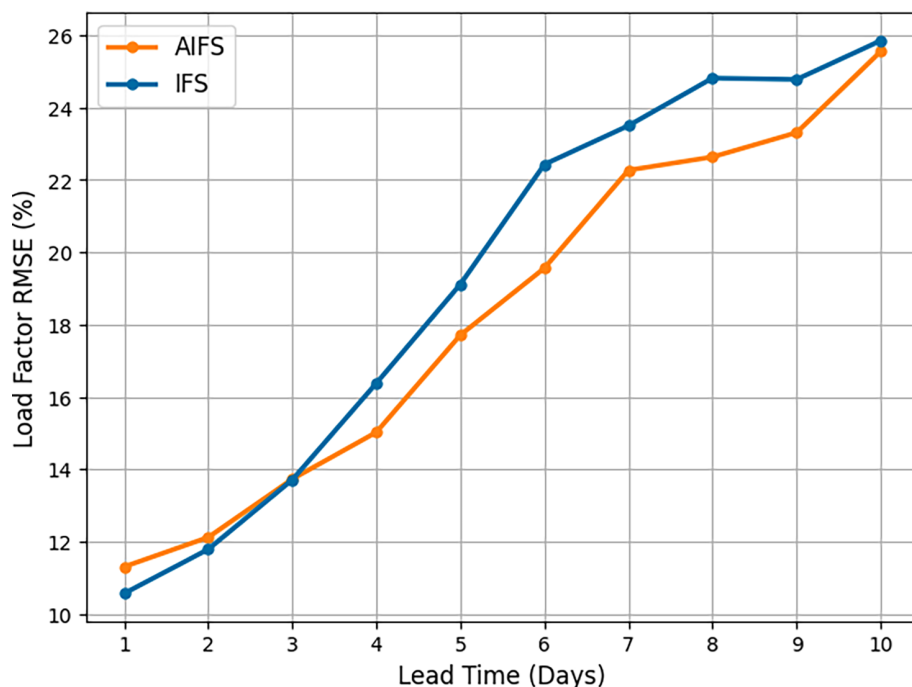


FIGURE 10 RMSE for the 2024 Kilgallioch wind power forecasts generated from our model (orange) and the IFS (blue) from March 2024 to November 2024. Actual power generation data downloaded from ELEXON was used as the 'ground truth'. The RMSE is averaged by each day in the forecast, that is, day 1 is the average of all the 6, 12, 18 and 24-h steps.

based approach extends the time for which the forecast is useful compared to the IFS, reinforcing the viability of our model for operational use.

3.4 | Spatial and temporal coherence

A common concern for AI models with spatial and temporal data is whether they possess coherence between local predictions. To test this, we isolated two wind farms in the United Kingdom, namely Kilgallioch in west Scotland and Hornsea Project Two, an offshore farm located east of the United Kingdom. We chose an initialisation date in 2024 when a low-pressure system tracked across the United Kingdom from west to east. Figure 11 shows a 5-day forecast for wind speed and power generation initialised on 1 July 2024 at 00Z at both farms using our model and the IFS outputs, and the actual values from the ECMWF analysis. We found that the AIFS and IFS predictions at both farms accurately capture variations from the wind current, suggesting the models exhibit spatial awareness. Additionally, the predictions for both models at Hornsea lagged behind the wind and power at Kilgallioch, capturing the time taken for the winds to travel across the United Kingdom, showing that temporal information is processed correctly.

A limitation of AI models is their tendency to 'smooth' predictions and their inability to capture the fine-grained behaviour of variables, particularly at longer lead times. Table 2 displays the average change in the predicted wind speed between successive time steps at

Kilgallioch and Hornsea Project Two across all forecasts in 2024 produced by our model and the IFS, and the 'actual' change from the ECMWF analysis. We found that our model displayed greater smoothing in the predictions than the IFS, indicated by the lower average change in wind speed. A model with increased parameters and longer training times could reduce this issue; however, we were fundamentally constrained by the performance of the AIFS, so it would likely produce minor improvements. Notably, when comparing the average change from the ECMWF analysis, we found that the IFS model also exhibits smoothing behaviour, suggesting that numerical approaches struggle to capture the variations accurately.

3.5 | Computational cost

An advantage of training a model that produces forecasts by leveraging an existing AI model is the reduced computational cost during inference and training. Our model processes a readily available AIFS forecast with a lead time of up to 15 days, where each input with the static variables added is 10.39 MB in size, producing an output size of 0.42 MB per time step. The relatively small number of parameters in our model results in fast inference speeds of 3.4 ms for a single time step on an entry-level Nvidia 3060 GPU. Additionally, we trained our model on ERA5 data from 2010, as opposed to the 1979–2020 range commonly used with other AI weather prediction models, resulting in significantly reduced training times

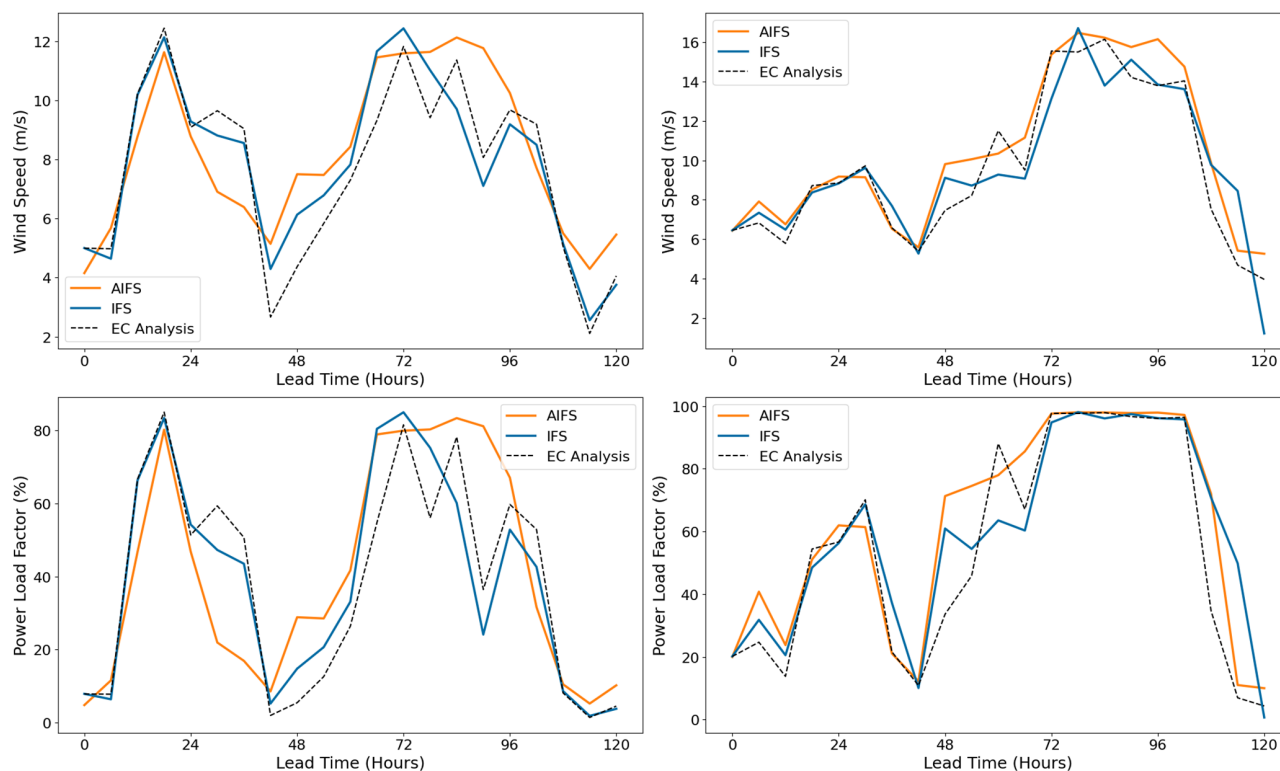


FIGURE 11 Plots showing the wind speed at hub height (top) and power load factor (bottom) for the AIFS (blue) and IFS (orange) 5-day forecast initialised at 2024-07-01 00Z. The ECMWF analysis (black dashed) at each time step is shown as the 'ground truth'. The time series are shown for the Kilgallioch (left) onshore wind farm and the Hornsea Project Two (right) offshore farm to illustrate the temporal and spatial coherence of the models.

TABLE 2 Table describing the average change in wind speed between successive time steps for each AIFS and IFS forecast in the 2024 dataset.

Data source	Kilgallioch (m/s)	Hornsea (m/s)
AIFS	1.46	1.69
IFS	1.76	2.13
ECMWF analysis	2.20	2.66

Note: The AIFS predictions are produced using our model with the AIFS forecast as inputs. The ECMWF analysis values are shown as the 'ground truth' for the models. A lower average change indicates more smoothing in the forecast values.

and a lower memory footprint. The reduced spatial input of 181×301 over a region of interest also resulted in fewer computational resources used. We also explored the use of mixed precision training to decrease the memory consumption of activations and improve training speed; however, this produced poorer predictive performance. Whilst AI-based models use fewer computational resources than NWP counterparts when producing forecasts, we acknowledge the expense to produce the ERA5 dataset used for training and the initialisation files used by the AIFS that currently depend on data assimilation

performed by the IFS. The novel concept of using an operational AI model output to predict additional variables has significant implications for the accessibility and cost of training powerful models. Following our method, it is possible to predict additional geographical variables found in ERA5 and other geographical datasets and across other domains where existing AI models do not provide the desired output.

4 | CONCLUSIONS

In this study, we have presented a novel approach for forecasting weather variables not available with current state-of-the-art AI models. By training a CNN on ERA5 data using a subset of variables found in the AIFS outputs, we created a model that can accurately predict the instantaneous 100-m wind speed with an average RMSE of 0.18 m/s. In comparison to the wind profile power law approximation for 100-m wind speed, which had an average RMSE of 0.63 m/s, our model was significantly more accurate. Using the AIFS forecasts from March 2024 to November 2024 as inputs to our model, we produced 10-day forecasts for the 100-m wind speed components.

We demonstrated that our results outperform the IFS forecast using ECMWF analysis data as the 'ground truth' at lead times of 3 days or greater while requiring a fraction of the computational cost.

We noted a bias in our predictions. There are two main differences between the ERA5 training data and the ECMWF analysis validation data that could cause this bias: the ERA5 data is generated using an earlier ECMWF cycle, and the analysis goes through an assimilation step. This bias is potentially leads to worse performance at the earlier lead times, where the majority of the error is from the bias, while at later lead times, the flow starts to dominate, making the bias a smaller comparative error.

We applied our 100-m wind speed forecast to predict the power generated by onshore and offshore wind farms across the United Kingdom. Similarly to the wind speed analysis, we compared our forecast to the IFS output and found that our model performs better on day 3 and later for both onshore and offshore farms. While our model demonstrated greater performance on later forecast days, we observed higher offshore errors and a drift in the wind speed distribution, indicating areas for further improvement. Additionally, we isolated a single wind farm to compare our wind power forecast to the actual UK grid generation and found that our model provided improvements over the IFS, particularly at later lead times.

We also address some common concerns with AI models in weather forecasting, in particular, the spatial and temporal coherence and the over-smoothing of predictions. By considering two wind farms located on the east and west sides of the United Kingdom, we demonstrated our model's awareness of the spatial and temporal relationships between its predictions. Despite our model's accuracy in capturing the correlations in the data, it exhibited over-smoothing in its predictions compared to the IFS forecast and the ECMWF analysis, as expected, emphasising a disadvantage of using AI-based methods. Nonetheless, we have produced forecasts that rival NWP approaches while using significantly less computational resources and presented a novel solution for predicting weather variables not available in current AI models.

AUTHOR CONTRIBUTIONS

Lucas Hardy: Methodology; software; formal analysis; writing – original draft; visualization. **Isla Finney:** Conceptualization; methodology; writing – review and editing; supervision.

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information (2024). ECMWF analysis, AIFS and IFS plots contain modified data from the European Centre for Medium-Range Weather Forecasts (ECMWF). Source www.ecmwf.int. This ECMWF data is published under a Creative Commons Attribution 4.0 International (CC BY 4.0) license <https://creativecommons.org/licenses/by/4.0/>. ECMWF does not accept any liability whatsoever for any error or omission in the data, their availability, or for any loss or damage arising from their use. BMRS data © Elexon Limited copyright and database right 2024, used under the BMRS open data license. <https://www.elexon.co.uk/data/balancing-mechanism-reporting-agent/copyright-licence-bmrs-data/>. Data from fig. 2 taken from Bloomfield et al. (2022) under CC 4.0 license.

CONFLICT OF INTEREST STATEMENT

Lucas Hardy and Isla Finney are employees of Lake Street Consulting Ltd.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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